

Abstract

Polyureas containing the diacyl hydrazine moiety in each repeating unit were synthesized via step-growth polymerization between 4-aminobenzohydrazide and hexamethylene diisocyanate at room temperature. The reactions were carried out in N, N-dimethylformamide (DMF) or N-methyl-2-pyrrolidone (NMP), with or without triethylamine as a catalyst. Formation of the polyurea was confirmed by IR and NMR spectroscopic analyses. The resulting polyureas exhibited oxidative degradability in calcium hypochlorite solution, yielding the corresponding dicarboxylic acid and evolving nitrogen gas (N₂), as verified by IR spectra. TGA and DSC analyses indicated that the polyurea synthesized using NMP is thermally stable up to **260 °C**, with a decomposition temperature (*T_d*) of **380.397 °C**; the polyurea synthesized using DMF is thermally stable up to **300 °C**, with a glass transition temperature (*T_g*) of **223.4 °C**, and was inert toward atmospheric oxygen. The materials showed an exceptional thin film-forming ability, with the films also being oxidatively degradable. Due to extensive intermolecular hydrogen bonding, the polymers were insoluble in most common organic solvents, dissolving only in NMP, DMF, and DMSO. Owing to their unique combination of thermal stability, controlled degradability, and film-forming properties, these polyureas hold significant potential for advanced research and industrial applications.